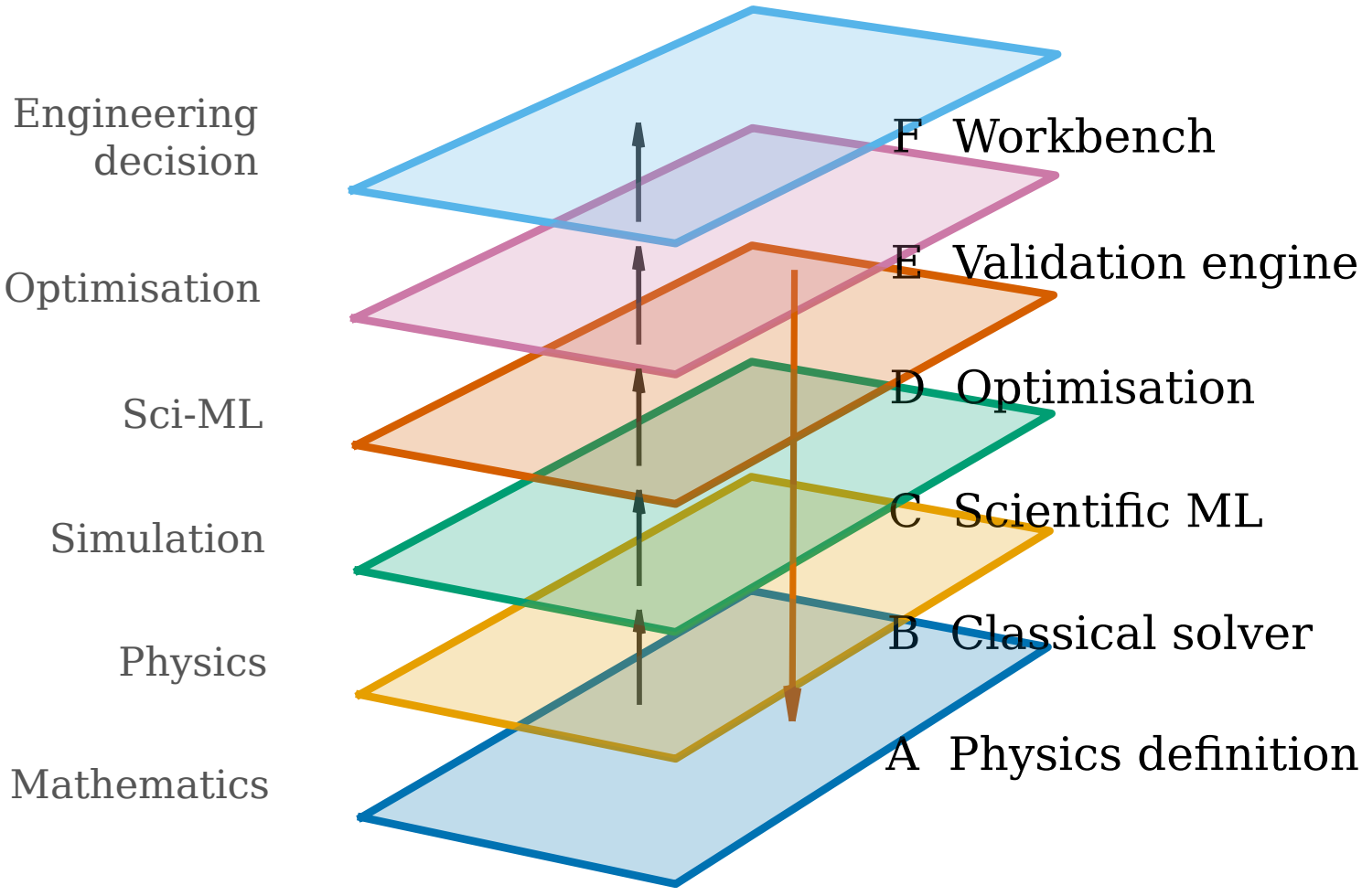
Schematic (not data): layered architecture and research pipeline



orange arrow: validation gate (trust / fallback)